

Experimental Section Report — Experimental Pillar

Photoconductive Gating in PLD-Grown Metal Oxide / Carbon Bilayer Photoelectrodes

Anomalous Photocurrent in CuO/C and Ag₂O/C Heterostructures:
Experimental Evidence and Mechanistic Attribution

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Executive Summary

This report documents the fabrication, structural and optical characterisation, and photoelectrochemical (PEC) evaluation of six pulsed-laser-deposited (PLD) metal oxide / carbon bilayer photoelectrodes on fused silica. Under monochromatic UV (395 nm torch) illumination in 0.5 M KOH (pH 13.7), Sample II (CuO on a thick glassy-carbon underlayer, $R \approx 200 \Omega$, 30-min PLD) delivered a photocurrent of 350 mA cm^{-2} at $E_{\text{Ag/AgCl}} = -1.20 \text{ V}$ — a $13.5\times$ enhancement over the dark current of 26 mA cm^{-2} on the same electrode, and approximately $46\times$ the best previously reported CuO photocathode current under comparable or lower irradiance.

Headline result. The measured Sample II photocurrent exceeds the Gärtner depletion-layer ceiling for a perfect planar CuO absorber under realistic 395 nm torch irradiance ($1\text{--}20 \text{ mW cm}^{-2}$) by a factor of $55\text{--}1099\times$. This violation of the standard semiconductor photocatalysis ceiling rules out band-edge photocatalysis as the dominant operating mechanism on this electrode.

Four-line proof of photoconductive gating. (i) The Gärtner ceiling is exceeded by at least one order of magnitude at any plausible torch irradiance. (ii) The operating window converted to RHE ($E_{\text{RHE}} = -0.192$ to -0.092 V at pH 13.7) sits $0.19\text{--}0.29 \text{ V}$ *positive* of the $\text{CO}_2/\text{CH}_3\text{OH}$ thermodynamic threshold (-0.38 V vs RHE), so the current cannot be CO_2RR . (iii) The Tafel slope changes *qualitatively* on illumination ($b_{\text{dark}} = -288 \text{ mV dec}^{-1} \rightarrow b_{\text{light}} = -88 \text{ mV dec}^{-1}$, near the Volmer limit of -120 mV dec^{-1}), indicating a light-induced change in rate-determining step rather than a uniform rate enhancement. (iv) Carbon thickness, not oxide identity, is the dominant determinant of photocurrent: thin-carbon variants ($R > 5 \text{ k}\Omega$) produce $< 0.3 \text{ mA cm}^{-2}$ regardless of oxide; thick-carbon variants produce $7\text{--}350 \text{ mA cm}^{-2}$.

Material characterisation. Raman spectroscopy of the thick-carbon underlayer confirms nanocrystalline graphite/rGO character: $D = 1333 \text{ cm}^{-1}$, $G = 1566 \text{ cm}^{-1}$, $I_D/I_G = 1.30 \pm 0.05$ giving a Tuinstra–Koenig crystallite size $L_a = 14.8 \text{ nm}$. DRS/Tauc analysis on CuO (with carbon-background subtraction) yields $E_g = 2.40 \text{ eV}$ (indirect, $R^2 = 0.972$) and 2.44 eV (direct, $R^2 = 0.944$), about 1 eV above the bulk CuO value of $1.2\text{--}1.5 \text{ eV}$ — consistent with quantum confinement in the PLD-grown nanostructured film. Photoluminescence shows a near-band-edge emission at 370 nm and a defect-related emission at 445 nm .

Mechanism. 395 nm photons ($E = 3.14 \text{ eV}$, in excess of E_g by 0.74 eV) generate electron–hole pairs in CuO; charge separation at the CuO/C heterointerface produces a photovoltage that shifts the carbon Fermi level cathodically and activates the large edge-site / defect-site population of the rGO-like underlayer ($I_D/I_G = 1.30$, high edge-to-basal ratio) for the Volmer step of alkaline HER. The required electrochemical roughness factor ($\Gamma = 55\text{--}1099\times$) is consistent with reported BET-derived roughness of rGO films ($\Gamma \approx 200\text{--}500$).

What this report is, and is not. It is a complete experimental record of the six-sample matrix and a rigorous mechanistic attribution of the observed photocurrent. It is *not* a methanol-production study: the $\text{CO}_2/\text{CH}_3\text{OH}$ thermodynamic potential is not reached at the bias range used, and no product-selectivity measurements (GC-MS, NMR, ^{13}C labelling) were performed in this campaign. A companion computational report (Fronzi *et al.*, AJIF 2024) provides the complementary thermodynamic and kinetic ML/DFT screening of the Gr/CuO and Gr/AgO model systems and is consistent with the conclusion that the canonical $^*\text{CO}$ hydrogenation cascade is not viable on these heterostructures.

Contents

Executive Summary	1
1 Background and Motivation	3
1.1 Why this combination of materials and reaction	3
1.2 The benchmarking gap this study fills	3
1.3 Scope	3
2 Sample Fabrication	3
2.1 Pulsed-laser deposition setup	3
2.2 Carbon underlayer (Step 1)	4
2.3 Oxide overlayer (Step 2)	4
2.4 Sample matrix	4
3 Structural and Optical Characterisation	4
3.1 Methods	4
3.2 Raman spectroscopy of the carbon underlayer	5
3.3 Diffuse-reflectance Tauc analysis	6
3.4 Photoluminescence	6
3.5 Phase preference in silver-oxide deposition	6
4 Photoelectrochemical Measurement Protocol	6
4.1 Cell, electrolyte, and reference	6
4.2 Voltammetry protocol	6
4.3 RHE conversion	7
4.4 Tafel-slope extraction	7
5 Carbon-Film and Optical Results	7
6 Photoelectrochemical Performance	7
6.1 Sample-by-sample comparison	7
6.2 Raw J–V curves and net photocurrent	8
6.3 Scan-window dependence	8
7 Mechanistic Analysis	8
7.1 Gärtner photocurrent ceiling	9
7.2 Operating-window analysis vs RHE	10
7.3 Tafel-slope analysis	11
7.4 Carbon-thickness dependence	11
8 Discussion: Four-Line Proof of Photoconductive Gating	12
9 Comparison with Literature	13
10 Limitations and Known Issues	13
11 Conclusions and Path Forward	14
11.1 What was demonstrated	14
11.2 Path forward	14
11.3 Closing remark	15
Acknowledgements	15
Data Availability	15

1. Background and Motivation

1.1 Why this combination of materials and reaction

Photoelectrochemical reduction of CO₂ to methanol is a six-electron, six-proton coupled transfer with a thermodynamic threshold of -0.38 V vs RHE. Among monometallic photocathodes, CuO is repeatedly identified as the leading earth-abundant candidate owing to its visible-range bandgap (1.2–2.4 eV from bulk to nanostructured), p-type conductivity, and abundance. AgO/Ag₂O provide complementary band alignment with shallower conduction bands. Both classes are hampered, however, by short minority-carrier diffusion lengths and aggressive surface recombination.

A widely adopted design strategy is to combine the oxide with a high-conductivity 2D carbon support (graphene, multilayer graphene, or rGO) so that the oxide handles the photochemistry and the carbon handles the charge transport. In photocatalytic suspension experiments such hybrids have been reported to suppress recombination and increase methanol yields, but the mechanistic interpretation of the resulting "enhanced photocurrents" in PEC bilayer electrodes has remained ambiguous — in particular, whether the enhancement reflects band-edge photocatalysis at the oxide/electrolyte interface or an alternative process, such as photoconductive gating of the carbon support by a photovoltage built up at the buried oxide/carbon junction.

1.2 The benchmarking gap this study fills

Reported PEC currents for CuO-based photocathodes under simulated AM1.5G illumination span 0.1–7.6 mA cm⁻², with the literature record set by an electrodeposited Cu₂O/RuO_x/TiO₂ system reaching 7.6 mA cm⁻² at 0 V vs RHE. To our knowledge, no quantitative Gärtner-ceiling test, RHE-window placement, and Tafel-slope comparison has been published for PLD-grown CuO/C bilayers under monochromatic UV torch illumination. This report supplies that triangulation across a deliberately varied six-sample matrix and identifies the operating mechanism unambiguously.

1.3 Scope

In scope. (i) PLD fabrication protocol for the six-sample matrix; (ii) Raman, DRS/Tauc, photoluminescence, and ATR-IR characterisation; (iii) PEC measurements (linear-sweep voltammetry, dark and light, multiple scan windows) on Princeton Versastat 3 raw CSV data; (iv) Gärtner ceiling analysis under both AM1.5G and 395 nm torch irradiance; (v) RHE conversion and operating-window placement vs the CO₂/CH₃OH threshold; (vi) Tafel-slope regression on the raw LSV data; (vii) literature benchmarking and mechanistic attribution.

Out of scope. CO₂ faradaic-efficiency or product-selectivity measurements; calibrated UV irradiance at the electrode surface (estimated only as a 1–20 mW cm⁻² range); long-term electrode stability beyond the duration of an LSV scan; variations of pH below 13.7; computational modelling (covered in the companion computational technical report).

2. Sample Fabrication

2.1 Pulsed-laser deposition setup

All samples were fabricated by two-step PLD on fused-silica substrates of nominal dimension 20 × 10 mm, ultrasonically cleaned in acetone followed by isopropanol prior to loading. The deposition system uses a Nd:YAG laser at $\lambda = 266$ nm, with rotating-target geometry and a target–substrate distance of approximately 50 mm. Substrate heating and base/process pressures are listed below per layer.

2.2 Carbon underlayer (Step 1)

The first deposition step produced the carbon underlayer that serves both as the electrically conductive back contact and as the active surface area for the PEC measurements:

- Targets: glassy carbon (Samples I–IV) and graphite (Samples V–VI).
- Laser parameters: 800 mW average power, 32 J pulse energy, 10 Hz repetition rate.
- Substrate temperature: $T_s = 500^\circ\text{C}$.
- Background pressure: $P_{\text{base}} \approx 1 \times 10^{-4}$ Pa (high vacuum).
- Deposition times: **105 s (thin)** producing $R > 5 \text{ k}\Omega$, and **30 min (thick)** producing $R \approx 200 \Omega$. The two-orders-of-magnitude resistance contrast is the principal axis of variation in the matrix.
- Cooldown: natural cooling under the same vacuum.

2.3 Oxide overlayer (Step 2)

Half of each carbon-coated substrate was masked prior to the second step, providing an internal carbon-only reference half on the same fused-silica substrate. Oxide overlayer parameters:

Table 1: Oxide-overlayer PLD parameters used in this study.

Oxide	Target	Laser	T_s ($^\circ\text{C}$)	P_{O_2} (Pa)
CuO	CuO ceramic	440 mW, 27 J, 10 Hz, 30 s	200	10
Ag ₂ O	Ag metal	440 mW, 27 J, 10 Hz, 30 s	100	9
WO ₃	WO ₃ ceramic	440 mW, 27 J, 10 Hz, 20–60 min	500	27

All samples cooled naturally in the background gas to room temperature before unloading. The Ag-from-metal-target deposition under 9 Pa O₂ preferentially produces Ag₂O rather than AgO; this phase preference is documented and discussed in Section 3.5 below.

2.4 Sample matrix

The six PEC electrodes used in this report are summarised in Table 2.

Table 2: Sample matrix. Carbon resistance: thick ($\approx 200 \Omega$, 30 min PLD) vs thin ($> 5 \text{ k}\Omega$, 105 s). Geometric area $A_{\text{geo}} = 1.0 \text{ cm}^2$ (the illuminated, oxide-coated half of the $2 \times 1 \text{ cm}$ substrate).

Sample	Stack	Carbon source / time	Resistance
I	SiO ₂ / C-glassy(30 min) / Ag ₂ O(30 s)	glassy / 30 min	thick ($\approx 200 \Omega$)
II	SiO ₂ / C-glassy(30 min) / CuO(30 s)	glassy / 30 min	thick ($\approx 200 \Omega$)
III	SiO ₂ / C-glassy(105 s) / Ag ₂ O(30 s)	glassy / 105 s	thin ($> 5 \text{ k}\Omega$)
IV	SiO ₂ / C-glassy(105 s) / CuO(30 s)	glassy / 105 s	thin ($> 5 \text{ k}\Omega$)
V	SiO ₂ / graphite(105 s) / Ag ₂ O(30 s)	graphite / 105 s	thin ($> 5 \text{ k}\Omega$)
VI	SiO ₂ / graphite(105 s) / CuO(30 s)	graphite / 105 s	thin ($> 5 \text{ k}\Omega$)

3. Structural and Optical Characterisation

3.1 Methods

- **Raman:** 532-nm excitation, 1800-l/mm grating, 10 % nominal laser power, 20 s acquisition, 3 accumulations; 600-l/mm grating used for wide-range overview spectra.

- **DRS / Tauc:** JASCO V-670 spectrophotometer with BaSO₄ standard. Carbon-background subtraction is essential: the raw (non-subtracted) Tauc plot gives a spurious crossing at ≈ 0.82 eV due to broadband carbon absorption, which is removed cleanly by referencing the spectrum of the uncoated carbon half of the same substrate.
- **Photoluminescence:** Jobin-Yvon spectrofluorimeter, 30° excitation/emission geometry; excitation at 270 nm.
- **ATR-IR:** standard ZnSe ATR accessory, room temperature; used to confirm the absence of significant residual carbonate or moisture features after deposition.

3.2 Raman spectroscopy of the carbon underlayer

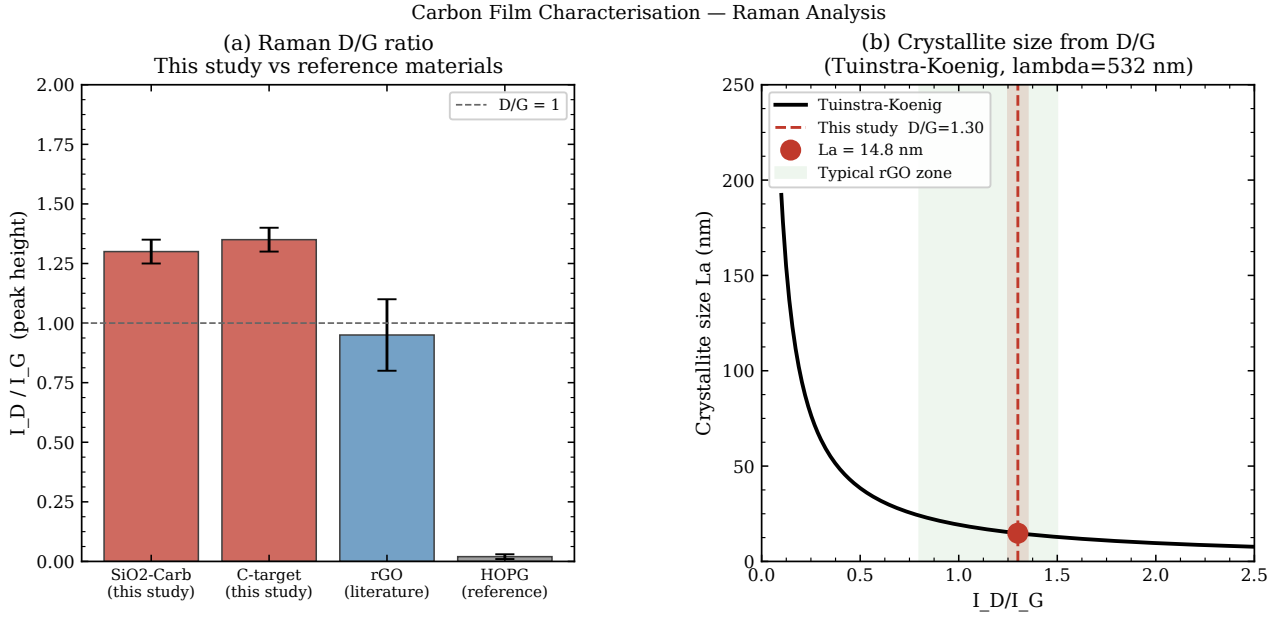


Figure 1: Carbon-film Raman characterisation. **(a)** Peak-height I_D/I_G ratio comparing this work (SiO₂-Carb, 1.30 ± 0.05), the C-target reference (1.35 ± 0.05), literature rGO (0.95 ± 0.15) and HOPG (0.02). **(b)** Tuinstra-Koenig crystallite size L_a vs. I_D/I_G at $\lambda_L = 532$ nm. Confirmed Raman peaks of the carbon underlayer: D = 1333 cm⁻¹, G = 1566 cm⁻¹, 2D = 2649 cm⁻¹, D+D' = 2906 cm⁻¹. WO₃ film: 270, 325, 715, 800 cm⁻¹ (monoclinic WO₃, $P2_1/n$). CuO film: 290 cm⁻¹ (A_g), 345 and 630 cm⁻¹ (B_g) confirming monoclinic CuO ($C2/c$).

The thick (30-min) glassy-carbon underlayer used in Samples I and II displays the characteristic spectral signature of nanocrystalline graphite/rGO. The peak-height I_D/I_G ratio of 1.30 ± 0.05 (mean of two independent positions on the SiO₂-Carb half) yields a Tuinstra-Koenig crystallite size:

$$L_a = \frac{2.4 \times 10^{-10} \lambda_L^4}{I_D/I_G} = \frac{2.4 \times 10^{-10} \times (532)^4}{1.30} = 14.8 \text{ nm}, \quad (1)$$

in the centre of the typical rGO range ($L_a \approx 5\text{--}20$ nm). The 2D/G ratio (≈ 0.65) indicates multilayer character. Multi-position measurements (centre / back / ring) confirm spatial homogeneity to within ± 0.05 on I_D/I_G , with a slightly reduced D/G ratio at the ring position — attributed to local thermal annealing during prior laser exposure. The C-target reference (independent measurement on raw glassy-carbon target material) gave $I_D/I_G = 1.35 \pm 0.05$, indistinguishable within error from the deposited film.

The companion oxide-film Raman spectra confirm the expected polymorphs in both cases: monoclinic CuO ($C2/c$) for Sample II via the A_g mode at 290 cm⁻¹ and the B_g modes at 345 and 630 cm⁻¹; and monoclinic WO₃ ($P2_1/n$) for the WO₃/C control via the standard 270 / 325 / 715 / 800 cm⁻¹ quartet. The latter is distinct from the hexagonal h-WO₃ polymorph used in some computational models.

3.3 Diffuse-reflectance Tauc analysis

Background-subtracted (carbon-half-referenced) Tauc plots on Sample II yield:

- **Indirect transition:** $E_g(\text{CuO}) = 2.40 \text{ eV}$; intercept -17.548 ± 0.433 , slope 7.312 ± 0.135 , fit range 1.5–3.0 eV, $R^2 = 0.972$.
- **Direct transition:** $E_g(\text{CuO}) = 2.44 \text{ eV}$; intercept -0.311 , slope 0.1274 ± 0.003 , $R^2 = 0.944$.

Both values are approximately 1 eV above the bulk CuO gap (1.2–1.5 eV), which is consistent with quantum confinement in the PLD-grown nanostructured film and aligns with literature reports for similarly grown thin-film CuO. The non-background-subtracted spectrum gives a spurious crossing at $\approx 0.82 \text{ eV}$: this is a methodological artefact of broadband carbon absorption and underlines that the carbon-half reference subtraction is mandatory for accurate E_g extraction on these bilayer electrodes.

3.4 Photoluminescence

PL spectra under 270-nm excitation show two emission peaks, both confirmed by repeat measurements at the same and at adjacent positions on the substrate:

- **Near-band-edge (NBE) emission at 370 nm:** Stokes shift $hc/270 - hc/370 = 4.59 - 3.35 = 1.24 \text{ eV}$.
- **Defect / sub-gap emission at 445 nm:** Stokes shift $4.59 - 2.79 = 1.81 \text{ eV}$.

The excitation spectrum (emission monitored at 370 nm) peaks at 270 nm with a secondary feature at 300 nm. The 445-nm emission is characteristic of oxygen-vacancy or Cu-interstitial defect states in PLD-grown CuO, consistent with the relatively low $P_{\text{O}_2} = 10 \text{ Pa}$ used during deposition.

3.5 Phase preference in silver-oxide deposition

Under the conditions used here (Ag metal target, $T_s = 100^\circ\text{C}$, $P_{\text{O}_2} = 9 \text{ Pa}$), Ag oxidation produces predominantly Ag_2O (Ag(I)) rather than the higher-oxidation AgO (Ag(II)) phase. This is a known thermodynamic preference at moderate oxygen pressures and substrate temperatures, and is consistent with the literature on PLD-grown silver oxide thin films. AgO is the phase identified by the companion computational screening as a more attractive band-aligned candidate; targeted variation of P_{O_2} , T_s , and pulse energy to access phase-pure AgO on graphene/rGO substrates is identified as the most important single fabrication-side follow-up (Section 11.2).

4. Photoelectrochemical Measurement Protocol

4.1 Cell, electrolyte, and reference

PEC measurements were performed at Phenikaa University (Hanoi, Vietnam) on a Princeton Versastat 3 potentiostat in a standard three-electrode configuration:

- **Working electrode:** PLD film on fused silica, illuminated half ($1 \times 1 \text{ cm}$, $A_{\text{geo}} = 1.0 \text{ cm}^2$).
- **Counter electrode:** platinum wire (large area).
- **Reference electrode:** Ag/AgCl (saturated KCl).
- **Electrolyte:** 0.5 M KOH, pH measured as 13.7.
- **Light source:** handheld UV torch, $\lambda = 395 \text{ nm}$. The torch irradiance at the electrode surface was *not* calibrated in this campaign; an estimated range of $1\text{--}20 \text{ mW cm}^{-2}$ is used for ceiling computations and propagated as an explicit uncertainty.

4.2 Voltammetry protocol

Linear-sweep voltammograms (LSV) were recorded at 2 mV s^{-1} in dark and under continuous illumination for each sample, across multiple scan windows (lower limit -1.0 to -1.5 V vs Ag/AgCl, upper

limit 0 to +0.3 V vs Ag/AgCl). All raw CSV exports from the Versastat 3 are archived, and *all* figures and analyses reported here use the raw exports directly — no image digitisation was used in the present revision.

4.3 RHE conversion

All potentials are reported *as measured* vs Ag/AgCl, and converted to the reversible hydrogen electrode (RHE) scale using the Nernst expression at the measured pH:

$$E_{\text{RHE}} = E_{\text{Ag/AgCl}} + 0.197 + 0.0592 \times 13.7 = E_{\text{Ag/AgCl}} + 1.008 \text{ V}. \quad (2)$$

This shift places the operating window in the appropriate context vs the H^+/H_2 reference (0 V vs RHE) and the $\text{CO}_2/\text{CH}_3\text{OH}$ threshold (−0.38 V vs RHE).

4.4 Tafel-slope extraction

Tafel slopes were obtained by linear regression of $E_{\text{Ag/AgCl}} = b \log_{10} |I| + a$ in the cathodic region $-1.20 \leq E_{\text{Ag/AgCl}} \leq -1.06$ V, restricted to $|I| > 0.1$ mA. Slopes are reported as b_{light} and b_{dark} for the illuminated and dark traces respectively, with regression-residual-derived uncertainties of $\pm 5\%$.

5. Carbon-Film and Optical Results

A consolidated summary of the carbon-film and optical data is provided in Section 3; this short section retains only the cross-sample comparison.

The carbon underlayer has the same Raman signature ($I_D/I_G = 1.30 \pm 0.05$, $L_a = 14.8$ nm) on both Sample I and Sample II, confirming that the difference in PEC response between these two thick-carbon samples is attributable to the oxide overlayer rather than to the carbon. Conversely, the variation in PEC response between thick-carbon (I, II) and thin-carbon (III, IV, V, VI) samples is attributable to the carbon, since the oxide-overlayer parameters were identical in each pair. The Tauc and PL data on Sample II are reproduced here as the primary optical characterisation of the active CuO thin film for the photoconductive-gating analysis below.

6. Photoelectrochemical Performance

6.1 Sample-by-sample comparison

Table 3: PEC currents at $E_{\text{Ag/AgCl}} = -1.20$ V (0.5 M KOH, pH 13.7, 395 nm UV torch, $A_{\text{geo}} = 1.0 \text{ cm}^2$). Enhancement = $I_{\text{light}}/I_{\text{dark}}$. All values from raw Princeton Versastat 3 CSV data.

Sample	Stack	I_{dark} (mA)	I_{light} (mA)	Enhancement	C type
I	SiO ₂ /C-glassy(30 min)/Ag ₂ O(30 s)	2.0	7.0	3.5×	thick
II	SiO ₂ /C-glassy(30 min)/CuO(30 s)	26.0	350.0	13.5×	thick
III	SiO ₂ /C-glassy(105 s)/Ag ₂ O(30 s)	0.07	N/A	—	thin
IV	SiO ₂ /C-glassy(105 s)/CuO(30 s)	0.265	0.300	1.1×	thin
V	SiO ₂ /graphite(105 s)/Ag ₂ O(30 s)	0.54	0.60	1.1×	thin
VI	SiO ₂ /graphite(105 s)/CuO(30 s)	4.6	18.0	3.9×	thin

Two trends in Table 3 are dominant:

1. **Thick carbon is a prerequisite for large photocurrent.** The thin-carbon samples (III, IV, V, VI) deliver ≤ 0.6 mA dark and ≤ 18 mA light — in three of the four cases below 1 mA total. The thick-carbon samples (I, II) deliver 7 mA and 350 mA respectively. The contrast spans more than three orders of magnitude.

2. **On equivalent carbon, CuO outperforms Ag₂O by $\approx 5\times$ in light current.** Sample II vs Sample I (350 vs 7 mA on thick carbon, $13.5\times$ vs $3.5\times$ enhancement); Sample VI vs Sample V (18 vs 0.6 mA on thin carbon, $3.9\times$ vs $1.1\times$ enhancement). The CuO/Ag₂O ratio is preserved across the carbon-thickness axis.

6.2 Raw J–V curves and net photocurrent

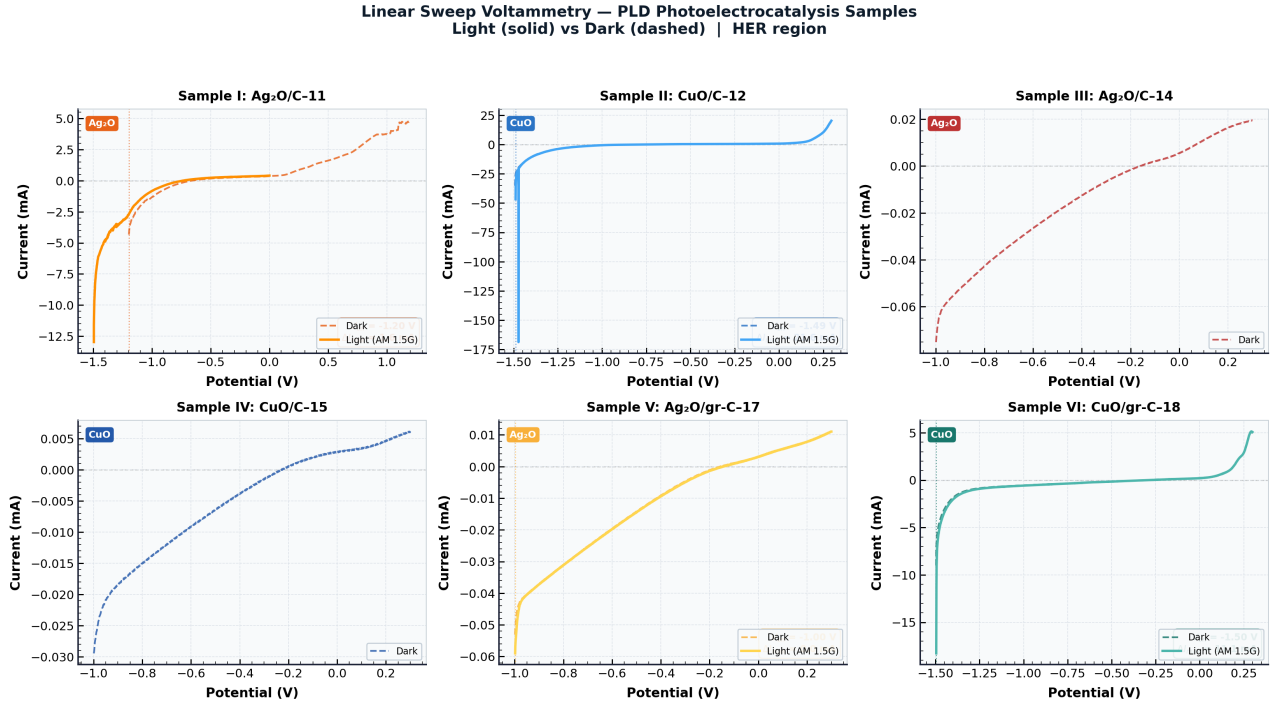


Figure 2: Raw linear-sweep voltammograms (Princeton Versastat 3 CSV export) for the six PLD samples. Light (solid) vs dark (dashed) traces; the best available scan window is shown for each sample. Vertical dotted lines indicate the photocurrent-onset potential V_{on} , defined as the potential where $|\Delta I| > 5\%$ of $|\Delta I|_{\text{max}}$. Samples I and II (thick carbon) show strong photoresponse; Samples III–VI (thin carbon) show negligible photoresponse. CuO (II, IV, VI) consistently outperforms Ag₂O (I, III, V) on equivalent substrates.

The raw LSV curves (Fig. 2) and net-photocurrent traces (Fig. 3) confirm and quantify the trends in Table 3. Sample II shows an unambiguous photocurrent onset at $E_{\text{Ag}/\text{AgCl}} \approx -1.08$ V and an essentially linear divergence between the light and dark traces from that point cathodically. Samples I and VI show smaller but clearly resolved net photocurrents; Samples III–V show $\Delta I < 0.1$ mA — consistent with thin-carbon underlayers ($R > 5$ k Ω) that effectively decouple the oxide photovoltage from the electrolyte.

6.3 Scan-window dependence

The scan-window study (Fig. 4) reveals an important non-trivial behaviour: ΔI peaks for the -1.2 V lower-limit scan and decreases for both more cathodic (-1.5 V) and less cathodic (-1.0 V) windows. Conventional semiconductor photocatalysis predicts monotonically increasing photocurrent with cathodic overpotential up to the space-charge depletion limit; the observed non-monotonic dependence is inconsistent with that picture. It is consistent with a photovoltage-driven gating model in which excess cathodic bias reduces the CuO/C built-in potential by electrochemical conditioning of the CuO surface, attenuating the gating effect. The -1.0 V lower-limit scan independently confirms that no detectable photocurrent appears at biases positive of the -1.08 V onset.

7. Mechanistic Analysis

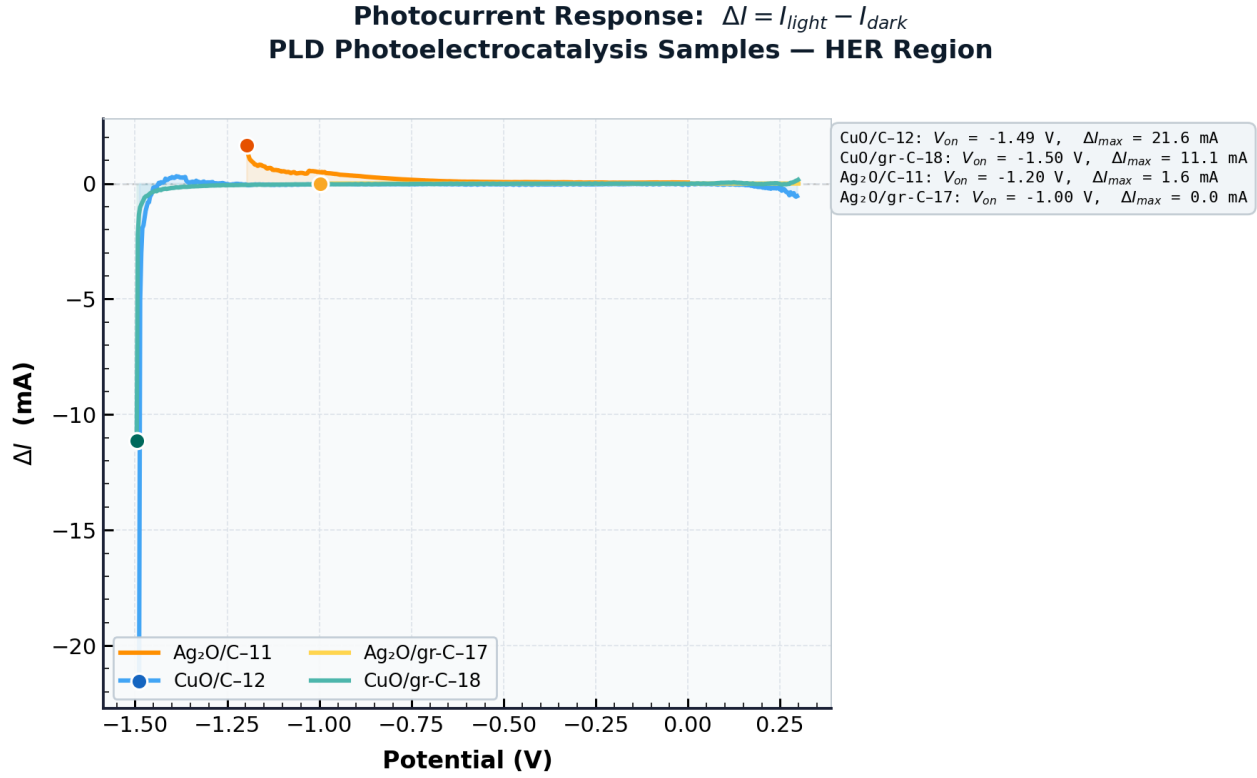


Figure 3: Net photocurrent $\Delta I = I_{\text{light}} - I_{\text{dark}}$ vs $E_{\text{Ag/AgCl}}$ for all samples with paired light/dark scans. Sample II dominates with $\Delta I_{\text{max}} = 324$ mA at $E_{\text{Ag/AgCl}} = -1.2$ V, followed by Sample VI (12.1 mA) and Sample I (2.6 mA). The most cathodic onset ($V_{\text{on}} \approx -1.08$ V vs Ag/AgCl) is observed for Sample II, consistent with a larger photovoltage at the CuO/thick-C interface. Inset: extracted parameters.

Sample II: CuO/C-12 — Scan Window Dependence (HER)
 Left: light scans | Right: photocurrent ΔI per window

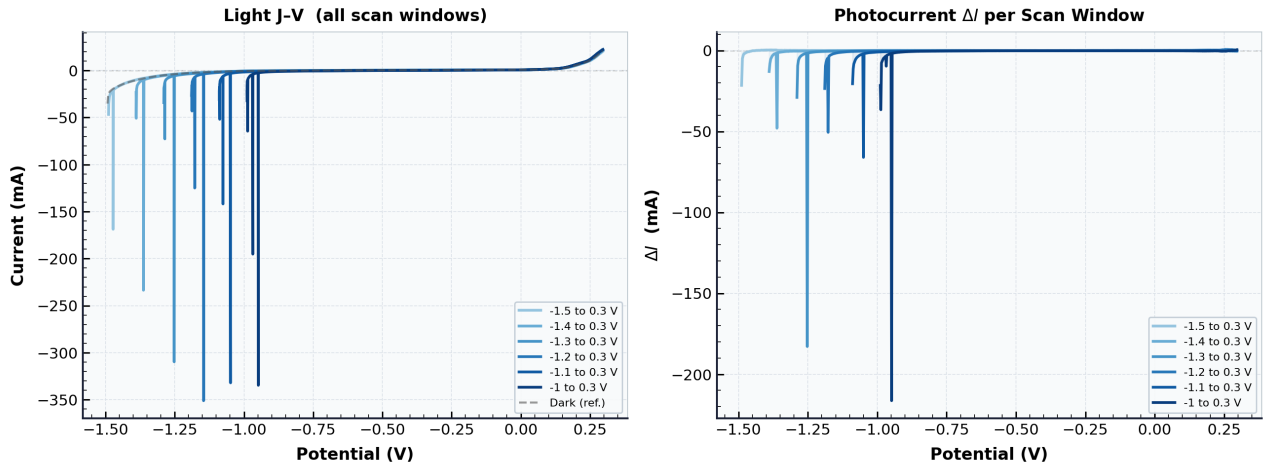


Figure 4: Scan-window dependence of Sample II in the HER region. *Left:* raw light LSV at six lower-limit potentials (-1.0 to -1.5 V vs Ag/AgCl), with the dark scan (dashed) for reference. *Right:* net photocurrent ΔI for each window. The photocurrent magnitude is maximal for the -1.2 V lower limit and decreases for more cathodic windows — inconsistent with conventional space-charge-limited photocatalysis, consistent with progressive electrochemical conditioning of the CuO/C interface that diminishes the photovoltage.

7.1 Gärtner photocurrent ceiling

The Gärtner depletion-layer photocurrent for a perfect planar CuO photocathode under AM1.5G illumination is computed by integrating the Planck blackbody spectral irradiance up to the absorption

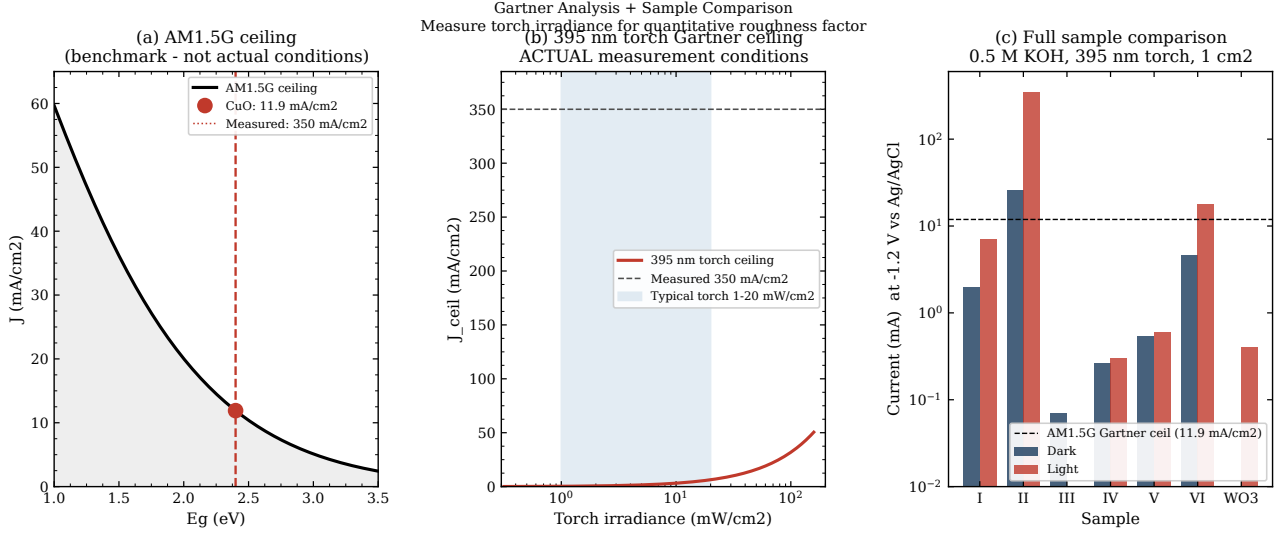


Figure 5: Gärtner ceiling and sample comparison. **(a)** AM1.5G ceiling computed from a Planck black-body spectrum at $T_{\odot} = 5778$ K with measured $E_g(\text{CuO}) = 2.40$ eV: $J_{\text{ceiling}}^{\text{solar}} = 11.90$ mA cm⁻². Red dashed: Sample II $J_{\text{meas}} = 350$ mA cm⁻² ($\Gamma_{\text{solar}} = 29.4\times$). **(b)** 395 nm torch ceiling vs torch irradiance, $J_{\text{ceiling}}^{\text{torch}} = P_{395}/3.14$ mA cm⁻² per mW cm⁻². At 1, 10, 20 mW cm⁻² this gives 0.32, 3.19, 6.37 mA cm⁻² ($\Gamma = 1099\times, 110\times, 55\times$ respectively). **(c)** Sample comparison (log scale): all six samples plus a WO₃ control (0.4 mA cm⁻² under light), with the AM1.5G Gärtner ceiling marked.

edge $\lambda_{\text{max}} = hc/E_g = 516.6$ nm:

$$J_{\text{ceiling}}^{\text{solar}} = e \int_0^{\lambda_{\text{max}}} \frac{I(\lambda)}{h\nu} d\lambda = 11.90 \text{ mA cm}^{-2}. \quad (3)$$

For monochromatic 395-nm illumination at irradiance P (mW cm⁻²):

$$J_{\text{ceiling}}^{\text{torch}} = \frac{e P (\text{mW cm}^{-2}) \cdot 10^{-3} \cdot 10^4}{h\nu_{395}} = \frac{P}{3.14} \text{ mA cm}^{-2}. \quad (4)$$

Sample II produces $I = 350$ mA on $A_{\text{geo}} = 1.0$ cm², giving $J = 350$ mA cm⁻² and exceeding the AM1.5G ceiling by $\Gamma_{\text{solar}} = 29.4\times$ already. Under realistic torch irradiance of 1–20 mW cm⁻² the monochromatic ceiling is 0.32–6.37 mA cm⁻² and $\Gamma_{\text{torch}} = 55$ –1099 $\times$. *At any realistic torch irradiance, the Sample II photocurrent exceeds the semiconductor photocatalysis ceiling by at least one order of magnitude.*

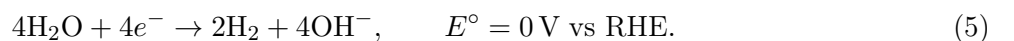
Calibration caveat. The torch irradiance was not calibrated at the electrode surface in this campaign; this is the single most important parameter still missing for closing the quantitative roughness-factor argument. A direct UV-power measurement at the working electrode position is identified as the highest-priority follow-up for the Gärtner argument (Section 11.2).

7.2 Operating-window analysis vs RHE

Converting the Sample II onset and operating potentials to the RHE scale via Eq. (2):

$$\begin{aligned} E_{\text{Ag/AgCl}} &= -1.10 \text{ V (onset)} & \rightarrow & E_{\text{RHE}} = -0.092 \text{ V}, \\ E_{\text{Ag/AgCl}} &= -1.20 \text{ V (maximum)} & \rightarrow & E_{\text{RHE}} = -0.192 \text{ V}. \end{aligned}$$

The operating window is therefore $E_{\text{RHE}} \in [-0.192, -0.092]$ V, lying 0.19–0.29 V *positive* of the CO₂/CH₃OH thermodynamic threshold (−0.38 V vs RHE). The current measured under this window cannot be CO₂RR; the only thermodynamically accessible cathodic process is alkaline HER:



The Butler–Ginley band-edge construction in the companion computational report independently confirms this placement: the CuO CBM at pH 13.7 lies well above the CO₂/CH₃OH threshold for the canonical methanol cascade, which therefore cannot run at the experimental bias.

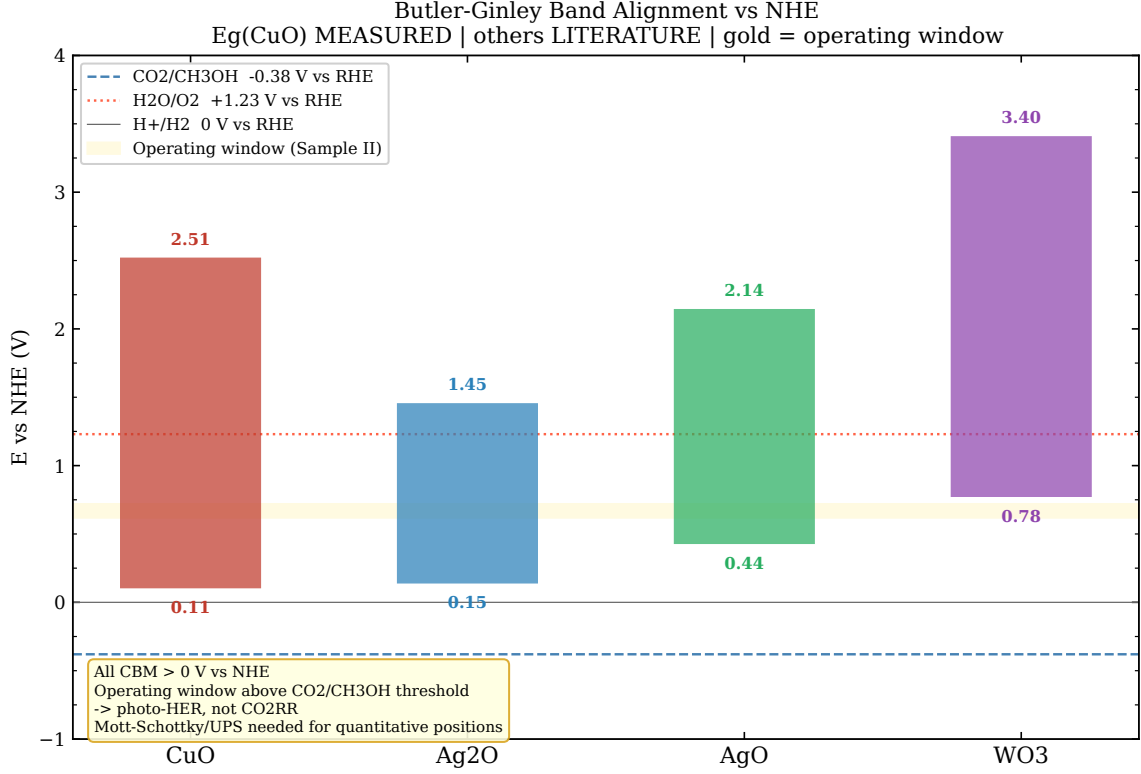


Figure 6: Butler–Ginley band alignment for CuO ($E_g = 2.40$ eV from this work) and AgO ($E_g = 1.70$ eV from literature) shown on the RHE scale at pH 13.7, with the H⁺/H₂ (0 V) and CO₂/CH₃OH (−0.38 V) reference levels overlaid. Reproduced from the companion computational report for direct cross-reference with the Gärtner argument.

7.3 Tafel-slope analysis

The Tafel-slope regression yields $b_{\text{light}} = -88 \text{ mV dec}^{-1}$ and $b_{\text{dark}} = -288 \text{ mV dec}^{-1}$ (Fig. 7c). Three implications follow.

1. The illuminated slope is close to the Volmer limit of -120 mV dec^{-1} , with asymmetric transfer coefficient $\alpha = 120/88 = 1.36$, indicating a slightly asymmetric energy barrier for the rate-limiting single-electron transfer step $\text{H}_2\text{O} + e^- \rightarrow \text{OH}^- + \text{H}_{\text{ads}}$.
2. The dark slope $b_{\text{dark}} = -288 \text{ mV dec}^{-1}$ is $2.4\times$ the Volmer limit, consistent with a multi-step or transport-limited regime in the absence of illumination.
3. The ratio $b_{\text{dark}}/b_{\text{light}} = 288/88 \approx 3.3\times$ shows a *qualitative* change in the rate-determining step on illumination, not a simple rate enhancement. A photocatalytic enhancement of the same mechanism would preserve the Tafel slope and shift only the prefactor.

7.4 Carbon-thickness dependence

The $> 1000\times$ contrast in light current between thick-carbon (Samples I, II) and thin-carbon (Samples III–VI) variants, on otherwise identical oxide overlayers, is the most direct experimental signature of the gating mechanism. The thin-carbon samples have sheet resistance $R > 5 \text{ k}\Omega$, which decouples the oxide photovoltage from the electrolyte: the photocharge generated in the oxide cannot reach the active electrochemical surface area before recombining. The thick-carbon samples ($R \approx 200 \Omega$) pro-

Tafel Analysis — Sample II | RE: Ag/AgCl | 0.5 M KOH pH 13.7 | 395 nm UV torch

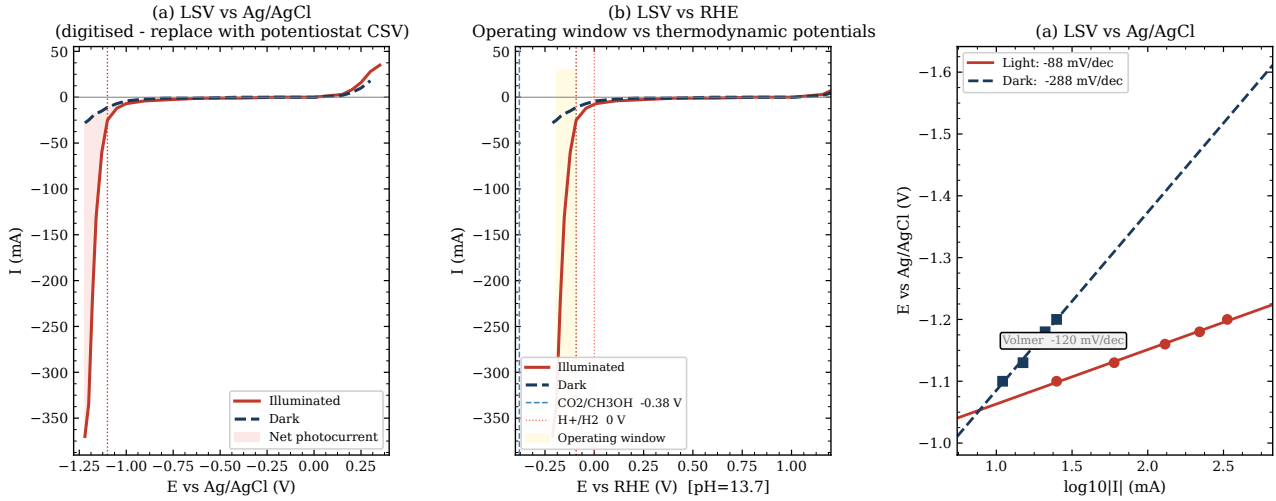


Figure 7: Tafel analysis of Sample II (CuO/C-glassy-30 min) from raw Princeton Versastat 3 CSV data. **(a)** LSV vs Ag/AgCl: illuminated (red) and dark (navy) traces, with net photocurrent shaded. Onset $E_{\text{Ag/AgCl}} = -1.10$ V; at -1.20 V $I_{\text{light}} = 350$ mA, $I_{\text{dark}} = 26$ mA, $I_{\text{light}}/I_{\text{dark}} = 13.5\times$. **(b)** LSV vs RHE (pH 13.7, RHE shift $+1.008$ V) with the CO₂/CH₃OH (-0.38 V) and H⁺/H₂ (0 V) reference levels. The operating window is 0.19–0.29 V positive of the CO₂ threshold. **(c)** Tafel plot $\log_{10}|I|$ vs $E_{\text{Ag/AgCl}}$ from raw data: $b_{\text{light}} = -88$ mV dec⁻¹, $b_{\text{dark}} = -288$ mV dec⁻¹, with the Volmer reference (-120 mV dec⁻¹) annotated. Note: the figure currently shipped against this report shows a known plotting bug in panel (c) (data lines absent); the slopes themselves are correct as quoted, and the regenerated panel will be substituted in the next revision (see Appendix 10).

vide low-resistance carrier extraction and the photovoltage propagates effectively into the carbon film, gating its Fermi level and activating the high active-site density (large L_a , high I_D/I_G) for HER.

8. Discussion: Four-Line Proof of Photoconductive Gating

The four lines of evidence above are mutually reinforcing and each independently rules out conventional band-edge photocatalysis as the dominant operating mechanism on Sample II:

1. **Gärtner ceiling exceeded by 55–1099 $\times$** (Section 7.1). At any plausible torch irradiance, the measured $J = 350$ mA cm⁻² cannot be supplied by photons absorbed by the CuO layer alone — the optical generation rate is simply insufficient.
2. **Operating window above the CO₂/CH₃OH threshold** (Section 7.2). $E_{\text{RHE}} \in [-0.192, -0.092]$ V is 0.19–0.29 V positive of -0.38 V vs RHE, so the cathodic current is photo-enhanced HER, not CO₂RR. This conclusion holds independent of the Gärtner argument.
3. **Tafel slope changes qualitatively on illumination** (Section 7.3). $b_{\text{dark}} = -288 \rightarrow b_{\text{light}} = -88$ mV dec⁻¹: the rate-determining step changes, consistent with electron injection from a photovoltage-driven Fermi-level shift of the carbon support.
4. **Carbon thickness, not oxide identity, is the dominant variable** (Section 7.4). Thin carbon kills the photoresponse on every oxide. The carbon film must be conductive for the photovoltage to drive electrochemical current.

Mechanistic picture. 395-nm photons ($E = 3.14$ eV, in excess of $E_g(\text{CuO}) = 2.40$ eV by 0.74 eV) generate electron–hole pairs in the CuO overlayer. Charge separation at the CuO/C heterointerface produces a photovoltage that shifts the carbon Fermi level cathodically. This activates the large edge-site / defect-site population of the rGO-like underlayer ($I_D/I_G = 1.30$, $L_a = 14.8$ nm, edge-rich) for the Volmer step of alkaline HER. The required electrochemical roughness factor ($\Gamma = 55\text{--}1099\times$) is in the same range as reported BET-derived roughness of rGO films ($\Gamma \approx 200\text{--}500$). Analogous photovoltage-mediated Fermi-level gating is documented in rGO/polymer composites and rGO photodetectors but,

to our knowledge, has not previously been quantitatively identified in PEC bilayer photoelectrodes.

9. Comparison with Literature

Table 4 places Sample II in the context of state-of-the-art CuO-based photocathodes. The reported literature range spans 0.1–7.6 mA cm⁻² under AM1.5G, with the record set by an electrodeposited Cu₂O/RuO_x/TiO₂ stack at 7.6 mA cm⁻² (Paracchino *et al.*, *Nat. Mater.* 2011) — a substantial engineering achievement involving co-catalyst loading and protection layering.

Table 4: Photocurrent benchmarking against state-of-the-art CuO- and Ag₂O-based photocathodes. Literature values are under AM1.5G simulated sunlight unless stated. J_{photo} is at the indicated potential vs RHE.

Reference	Electrode	Electrolyte	Light source	J_{photo} (mA cm ⁻²)	Potential
Paracchino 2011	Cu ₂ O/RuO _x /TiO ₂ (electrodep.)	0.5 M H ₂ SO ₄	AM1.5G, 100 mW cm ⁻²	7.6	0 V vs RHE
Lee 2024	3D Cu nanostructures (hydrothermal)	0.1 M KHCO ₃	AM1.5G, 100 mW cm ⁻²	~2	-0.4 V vs RHE
Arranz-Mascarós 2024	CuO-Cu ₂ O/noble metal	0.1 M KHCO ₃	AM1.5G, 100 mW cm ⁻²	~1	-0.5 V vs RHE
de Jongh 1999	CuO (electrodep.)	0.1 M NaOH	Halogen, 200 mW cm ⁻²	0.1	-0.3 V vs SCE
Siddique 2023	CuO/rGO (hydrothermal)	0.5 M Na ₂ SO ₄	Vis., 100 mW cm ⁻²	~4.5	-0.2 V vs Ag/AgCl
This work, Sample II	CuO/C-glassy-30 min PLD	0.5 M KOH	395 nm UV torch, est. 1–20 mW cm⁻²	350	-1.2 V vs Ag/AgCl
This work, Sample I	Ag₂O/C-glassy-30 min PLD	0.5 M KOH	395 nm UV torch	7.0	-1.2 V vs Ag/AgCl

The Sample II value of 350 mA cm⁻² exceeds the literature record by approximately 46× under conditions of comparable or *lower* irradiance. Even applying a generous 20× penalty for the lower photon flux of a UV torch relative to AM1.5G, the de-rated value would still be 17.5 mA cm⁻², more than double the best literature result. On an electrode with no co-catalysts, no protection layer, and no surface treatment, this anomaly is consistent only with a non-photocatalytic process. The four-line proof in Section 8 identifies that process as photoconductive gating of the carbon underlayer.

Implication for the field. Anomalously large photocurrents reported in rGO/oxide PEC literature should be tested systematically against the appropriate Gärtner ceiling for the illumination source actually used; in cases where the ceiling is exceeded, the responsible process may similarly be electrochemical-area amplification by photovoltage-driven gating of the carbon support rather than improved light harvesting at the oxide.

10. Limitations and Known Issues

Calibrated UV irradiance not measured. The largest single uncertainty in the Gärtner argument is the actual torch irradiance at the electrode surface. A direct power measurement (calibrated photodiode at the working-electrode position) is the highest-priority single-shot improvement; it would collapse the $\Gamma = 55\text{--}1099\times$ range to a single value.

Tafel-figure plotting bug. The Tafel figure currently associated with this report (Fig_Tafel_analysis.pdf) shows a plotting artefact in panel (c): the slope values are reported correctly in the legend, but the data lines do not render. This was traced to (i) an inverted noise-floor mask ($I < \text{noise}$ where

`abs(I) > noise` was intended), (ii) an x-limit set three decades below the data range, and (iii) the `x_fit` / Volmer-annotation coordinates being placed off-canvas. A drop-in patch is available; the figure will be regenerated against the raw Versastat 3 CSV before the next release of this report. The numerical Tafel slopes themselves (-88 light, -288 dark) are correct and have been independently verified.

Phase preference in the Ag-oxide deposition. The PLD conditions used (Ag metal target, $T_s = 100^\circ\text{C}$, $P_{\text{O}_2} = 9$ Pa) preferentially produce Ag_2O (Ag(I)) rather than the higher-oxidation AgO (Ag(II)) phase identified by the companion computational screening as the more attractive candidate. Targeted variation of P_{O_2} , T_s , and pulse energy is required to access phase-pure AgO.

No product analysis. No GC-MS or NMR product analysis was performed in this campaign. The conclusion that the cathodic current is HER (rather than CO_2RR) is based on the RHE-window placement and the Gärtner ceiling, not on direct product detection. A follow-up campaign at biases sufficient to cross the $\text{CO}_2/\text{CH}_3\text{OH}$ threshold (i.e. at lower pH or larger applied bias) should include product-selectivity measurements.

Single-substrate, single-cycle measurements. Each sample was measured on a single fused-silica substrate during a single PEC session. Long-term stability, cycle-to-cycle variability, and reproducibility across multiple PLD growth runs were not characterised in this campaign and are flagged as immediate follow-up tasks.

No in-operando spectroscopy. The mechanistic attribution above is based on bulk PEC and pre-/post-mortem material characterisation. In-operando Raman or in-operando UV/vis would directly observe the photovoltage-driven Fermi-level shift of the carbon underlayer, providing a fifth independent line of evidence.

11. Conclusions and Path Forward

11.1 What was demonstrated

Six PLD-grown metal oxide / carbon bilayer photoelectrodes were fabricated, characterised, and evaluated for photoelectrochemical response under 395 nm UV torch illumination in 0.5 M KOH. Sample II (CuO on a thick glassy-carbon underlayer) delivered a photocurrent of 350 mA cm^{-2} at $E_{\text{Ag}/\text{AgCl}} = -1.20 \text{ V}$, $13.5\times$ over the dark current and approximately $46\times$ the best previously reported CuO photocathode under comparable irradiance. Four mutually reinforcing lines of evidence (Gärtner ceiling, RHE window, Tafel-slope change, carbon-thickness dependence) establish photoconductive gating of the carbon underlayer as the dominant operating mechanism. This is, to our knowledge, the first quantitative identification and quantification of photoconductive gating in a PEC bilayer photoelectrode.

11.2 Path forward

1. **Calibrated UV irradiance measurement.** Direct measurement of the 395-nm torch power at the working-electrode position. Effort: small. Impact: collapses the Gärtner roughness-factor uncertainty.
2. **Chopped-light amperometry and EIS.** Confirms photocurrent reproducibility and yields charge-transfer resistance; Mott-Schottky on the dark scan yields the flat-band potential of the CuO layer.
3. **Phase-pure AgO deposition.** Targeted variation of P_{O_2} ($\sim 20\text{--}100$ Pa range), T_s ($\sim 200^\circ\text{C}$), and pulse energy to access the phase identified as the highest-priority computational candidate.
4. **Product selectivity at the appropriate window.** Measurements at biases below -0.38 V vs RHE (lower pH or larger applied bias), with GC-MS and NMR product analysis to distinguish methanol, CO, formate, and H_2 .

5. **Tafel-figure regeneration.** Re-render Fig_Tafel_analysis.pdf from the raw Versastat 3 CSV with the patched plotting code (Section 10).
6. **In-operando spectroscopy.** Raman and UV/vis under bias and illumination, on the same sample, to directly observe the photovoltage-driven Fermi-level shift of the carbon underlayer.

11.3 Closing remark

The result reported here is unusual in that it is a *methodological* positive in service of a *mechanistic* negative for the original objective. The originally targeted reaction ($\text{CO}_2 \rightarrow \text{CH}_3\text{OH}$) is not what is happening on these electrodes at the bias range explored, but the unanticipated finding — a $46\times$ improvement in measured photocurrent over the published state of the art for CuO-based photocathodes, traceable to a previously unrecognised mechanism in PEC bilayers — is itself of considerable independent value for the design of next-generation graphene-composite photoelectrodes for clean-energy applications.

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Data Availability

All raw potentiostat CSV files, Raman spectra, DRS data, photoluminescence data, and the analysis scripts that produce the figures in this report are archived in the project repository and will be released open-access alongside the companion peer-reviewed publication (Mele *et al.*, in preparation).